

Gujarat Technological University

Master of Computer Applications (MCA) — Semester 3

Course: Mobile Application Development (MC03094031)

Assignment: 30 Questions (7 Marks Each) — For GTU Examination Reference

Unit 1: Basics of Android

Q1. Explain the features of Android SDK and describe the tools used in Android application development. [7]

Q2. Describe the Android application life cycle with the help of a diagram, explaining each state and its callback methods. [7]

Q3. What is the AndroidManifest.xml file? Explain its role and key elements with an example. [7]

Q4. Explain the concept of externalizing resources in Android. Why is it important, and how is it implemented? [7]

Unit 2: User Interface in Android

Q5. Explain different types of layouts available in Android (LinearLayout, RelativeLayout, ConstraintLayout, FrameLayout, GridView) with examples. [7]

Q6. What are Fragments in Android? Explain their life cycle and how they differ from Activities. [7]

Q7. Explain Intents in Android. Differentiate between explicit and implicit intents with examples. [7]

Q8. What are Broadcast Receivers? Explain how to create and register a broadcast receiver with an example. [7]

Q9. Explain the concept of Adapters in Android. Describe how ArrayAdapter and BaseAdapter are used. [7]

Q10. Explain how to download and parse internet resources in Android using an example. [7]

Q11. Describe the Download Manager in Android and explain its use with a suitable example. [7]

Unit 3: Working with Data, Databases, and Background Processing

Q12. Explain Shared Preferences in Android. Write steps to store, retrieve, and modify data using Shared Preferences. [7]

Q13. Explain how to work with the file system in Android for saving and reading files. [7]

Q14. Explain SQLite database in Android. Describe the steps to perform CRUD operations using SQLiteDatabase and SQLiteOpenHelper. [7]

Q15. What are Content Providers? Explain how to create and use a custom Content Provider with an example. [7]

Q16. Explain Content Values and Cursor objects in Android with suitable examples. [7]

Q17. Describe native Android Content Providers with examples (e.g., Contacts, MediaStore). [7]

Q18. Explain Android Services. Differentiate between Started Service and Bound Service. [7]

Q19. Explain background threads and alarms in Android. How are they used to perform tasks without blocking the UI? [7]

Unit 4: Enhancing User Experience

Q20. Explain the Action Bar in Android. Describe how to create and use menus and action bar action items. [7]

Q21. Explain different types of Dialogs in Android (Alert Dialog, Custom Dialog) with an example. [7]

Q22. What are Notifications in Android? Explain the steps to create and display a notification. [7]

Q23. Explain how to design Android applications for different screen sizes and densities. [7]

Q24. Explain Text-to-Speech and Speech Recognition features in Android with implementation steps. [7]

Q25. Explain Animations in Android. Describe the different types of animations with examples. [7]

Unit 5: Audio, Video, Camera, Telephony, and SMS

Q26. Explain the steps to play audio and video files in an Android application. [7]

Q27. Explain how to use the camera in Android to capture pictures and record video, and add media to the media store. [7]

Q28. Explain how to send and receive SMS in Android with a suitable example. [7]

Unit 6 & 7: Publishing the App and Web Services

Q29. Explain the steps involved in signing and publishing an Android application on the Google Play Store. [7]

Q30. Explain how to consume a web service (JSON/XML) in Android with an example, including user authentication using a web service. [7]

Note: These questions are compiled from the full syllabus (Units 1–7) for exam-oriented practice. Students should also refer to their lab manual and reference books listed in the syllabus for practical understanding.